A monolithic implementation of a switched-current "wheel" amplifier for multi-channel Silicon Drift Detectors[1,2]

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Abstract

In this paper we present a new VLSI shaper amplifier suitable for multi-anode Silicon Drift Detectors, or other multi-element solid-state X-ray detection systems. The read-out scheme has been conceived and developed for demanding applications with synchrotron light sources, such as X-ray holography or EXAFS experiments, where both high count-rates and high energy resolutions are required. The circuit is of time-variant nature, accepts randomly distributed events and features: a finite-width (1.1μs) weight function, a low-level energy discrimination, a low power consumption (a few mW per channel) and a full compatibility for monolithic integration in CMOS technology. Its impulse response has a staircase-like shape, but the implemented weight function (which is in general different from the impulse response in time-variant systems) is quasi trapezoidal. A first prototype of the circuit is here described and the first experimental results are also reported.

I. Introduction

In the last decade many research groups worldwide have designed and developed state-of-the-art non-cryogenical systems for high-rate high-resolution X-ray spectroscopy [1-4]. Contextually new applications based on spectroscopic measurement techniques have been proposed in the fields of fundamental physics, archaeometry, molecular biology. Substantial advances have been achieved in the technological development of appropriate multichannel X-ray detectors. For example high-yield mass production of new-generation multi-anode Silicon Drift Detectors (SDD's) for X-ray holography is nowadays possible [5], and experiments requiring thousands of detection channels may be proposed to the scientific community. Besides, various solutions have been developed in order to deliver a compact front-end electronics and an adequate back-end digital circuitry. Recently, an innovative VLSI time-variant signal processor suitable for such applications, and featuring a narrow, quasi-optimal trapezoidal Weight Function (WF), has been proposed by our group [6]. Such new processor is adequate for implementations in arrays, and, despite its time-variant nature, is capable to correctly process the events even when they are distributed randomly along the time axis and no synchronism signal is available. Such processing circuit gathers the advantages of the classical continuous-time spectroscopy amplifiers (no synchronism required, very low level discrimination) and those of time-variant processors (finite width, flat topped optimal WF's), at the expense of some increased complexity in its topological structure. In this paper such new circuit, referenced as "wheel amplifier" due to its circular structure, is described and experimental results, obtained with a first complete prototype, are presented. The circuit has been layed out for the 1μm/5V JFET/CMOS technology of the Fraunhofer-Institut für Mikroelektronische Schaltungen und Systeme, Duisburg and is a candidate for a big experiment founded by the Eropean Community.

II. Overview of the Working Principles

In Fig.1 a single-channel processor is shown schematically. It consists of a common input preamplifier, a "wheel" of a given number of identical processing units (four in the implemented prototype) operated in sequential order,

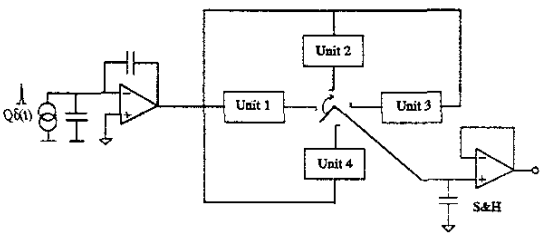

Figure 1: Principle of operation of the "wheel" amplifier: four staggered identical units, fed by a common input amplifier, are used in this schematic. The outputs of the units are sequentially sampled by a S&H.

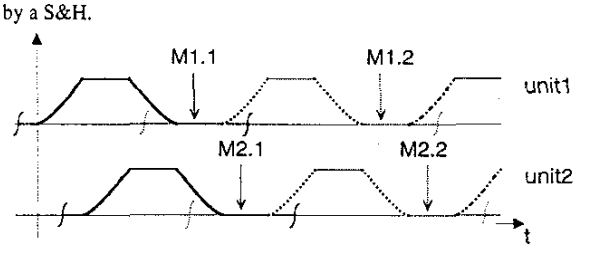

Figure 2: Weight functions ideally implemented by units 1 and 2. The weight functions of the other units are not reported in the figure but they must be staggered in a similar way. The rotating switch of Figure 1 is operated at the times of measurements *M1.1*, M2.1, M3.1, M4.1, *M1.2*, M2.2, ...

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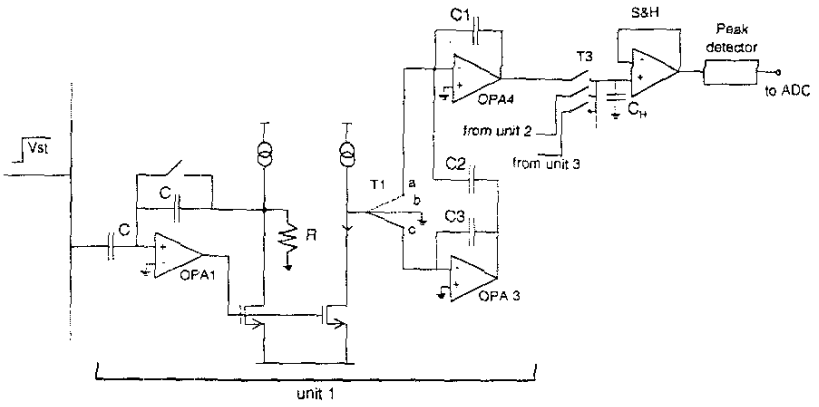

Figure 3: Simplified schematic of the "wheel" amplifier. An individual unit is fully depicted. It basically consists of a transconductor (OPA1) and two integration stages (OPA3, OPA4). The output is connected to a common Sample and Hold stage.

and a common Sample and Hold output stage. In Fig.2 the weight functions ideally provided by means of units 1 and 2, from the input current δ pulses to the S&H output, are sketched. Note that each unit provides actually a sequence of optimal flat topped weight functions (solid, dotted, and dashed dotted lines), each of which is referred to a sequence of corresponding measurement times (for example M1.1, M1.2, ... for unit 1). We recall that the "flatted cusp" is the minimum-noise finite-width filter for nuclear spectroscopy. The weight functions implemented by unit 2 (also reported in Fig.2) are staggered in such a way that their flat tops be contiguous or partially overlapped to those of unit 1. The weight functions of units 3 and 4 (not reported in the figure) have to be in turn staggered in a similar way. The rotating switch of Fig.1 will be operated so as to sequence the measurements in the order M1.1, M2.1, M3.1, M4.1, M1.2, M2.2, etc.

The fundamental point which is worth stressing here is that in this way the time axis may be fully covered by the flat tops provided by the weight functions of the various units. This is very important because it guarantees that wherever the input current signal occurs there is at least one unit which captures the event under its flat top, providing a correct measurement of the event itself. For a given input signal three units will provide a non-vanishing output signal: first the unit which sees the event under the right shoulder (see Figure 2) of its weight function, then the one which sees the signal under its flat top, eventually that one which sees the event under its left shoulder. The maximum value among these measurements will be that provided by the unit which is processing the event under its flat top, because it features the maximum gain. Therefore such an output signal will look comprehensively like a staircase, whose maximum is the correct measurement of the event. The good measurement may be hence obtained by means of a peak detector-stretcher (see Sect. C). For further details about the principles of operation of the "wheel amplifier" see also refs. [6].

III. THE PROPOSED IMPLEMENTATION

A. *Input preamplifier*

The first amplification stage of the proposed solution is a voltage preamplifier obtained by means of a coupling capacitor and a charge preamplifier. The requirement of a voltage preamplifier instead of a charge preamplifier was motivated by the need of using the present circuit with Silicon Drift Detector with integrated JFET, which is operated in a source follower configuration [5]. The design of this circuit is based on a previous solution used in CAMEX [7]. Particular care has been taken in the design of the input transistor (a n-channel JFET) for what concerns noise performances. The choice of the optimum value of the coupling capacitor (C_j = 5 pF) was derived from a compromise between the need of a short rise time (small values of C_j) and the need of negligible noise contribution introduced by the preamplifier (large values of C_j).

B. *Wheel structure*

In Fig.3 a simplified schematic of the implemented "wheel amplifier" is shown. The reset switches on C3 and C2 are not shown for the sake of simplicity. In the figure an individual unit is shown (referenced as unit 1). The step voltage signal V_{st} provided by the preamplifier is converted into a current by means of a resettable transconductor (OPA1, R, C), note that one per each unit is used. The trapezoidal weight functions are then constructed by properly switching the current provided by the transconductor by means of T1. First the current is integrated over C1 (position *a* of T1), then it is flown to ground (position *b*), eventually it is integrated over C3 (position *c*) whereas OPA4 acts as an inverting amplifier. It may be shown (see [6]) that by sequencing the operations in such a way a trapezoidal weight function results, provided that the two integration times are equal and C2=C3. After such switching sequence is accomplished the output voltages are sampled through switches T3, which actually implement the

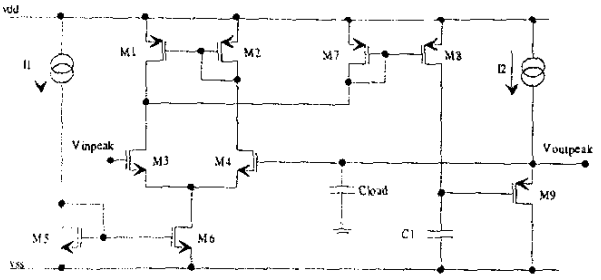

Figure 4: The Peak Stretcher circuit.

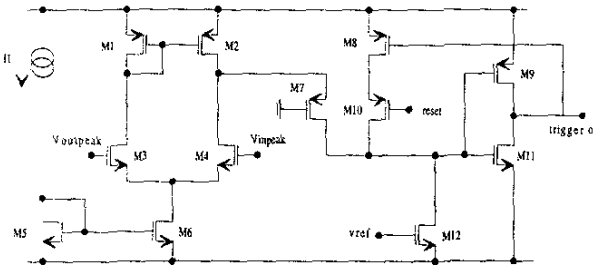

Figure 5: The Threshold Discriminator circuit.

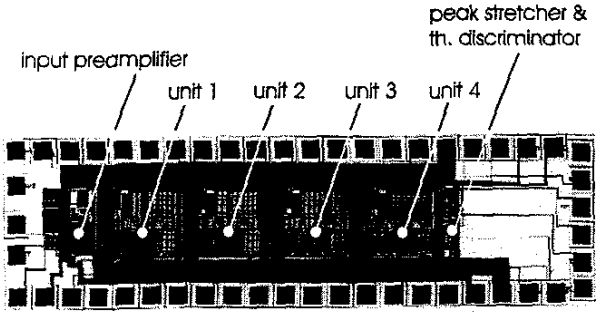

Figure 6: Layout of the first prototype of the circuit.

rotating switch of Fig.1. In order to guarantee a correct operation all the units should be identical. The uniformity of the gains of the units is related to technological parameters (accurateness in the ratio of on-chip resistor and capacitor values). With the used technology the expected disomogeneites are in the subpercent range, which is compatible with high resolution X-ray spectroscopy (up to a few tens of keV), where the spectral resolutions are typically in the some-percent range.

C. *Peak stretcher and threshold discriminator*

The output of the S&H (which samples sequentially the output of each wheel at the measurement time) is connected to a Peak Stretcher (PS) which has to maintain the maximum value of the "staircase" signal for the ADC conversion. The design of the PS, whose schematic is shown in Figure 4, is based on the topology proposed by Kruiskamp and Leenaerts [8], which uses a MOS current mirror in place of the more traditional rectifying diode. The circuit has been designed to track and hold a positive pulse as follows. As long as $V_{outpeak}$ is lower than V_{inpeak} (which corresponds to the S&H output) the capacitor C_1 is charged by the current delivered at the output of the differential stage and mirrored by means of M7 and M8 (tracking phase). As soon as V_{inpeak} becomes lower than $V_{outpeak}$, the current mirror (M7,M8) switches off and C_1 is no more charged (hold phase). C_2 stands for the input capacitance of the output driving buffer (not shown in figure). The choice of C_1 results as a compromise between the optimization of several circuit characteristics: closed-loop bandwidth, minimization of the charge injection when M8 is turned off, droop rate due to the quiescent current of M8 and physical size. Moreover, the time constant of the output follower, which depends on the output capacitive load C_{load} and on the transconductance of M9, has to be set short enough to guarantee a fast interruption of the charge of C1 at the beginning of the hold phase.

The Threshold Discriminator (TD), shown in Figure 5, provides a negative output trigger pulse in correspondence of the event occurence, in order to enable the ADC conversion. A reset signal from the ADC is expected when the conversion is over. The working principle of this circuit is the following. Let us suppose that the reset (active-high) is low, the trigger output is high and also that the peak detector is not in hold phase so that $V_{outpeak}$ is less or equal to V_{inpeak}. In this situation, no current flows through M7 and M12. Even if V_{ref} is higher than the threshold voltage, M12 is working on the origin of its I_D versus V_{DS} curves, so at $I_D = 0$. When the hold phase starts, current flows though M7 and M12. M12 goes out from its ohmic region to the saturation region, when the current arriving from the previous stage is equal to the saturation current corresponding to $V_{GS} = V_{ref}$. In this way the drain voltage of M12 is suddenly raised up. The output inverter (M9, M11) provides a negative output signal which turns on M8, placed in a positive loop, with the consequence that the trigger output signal is forced to lower further. The positive reset signal from the ADC, applied both to the PD and to M10, stops again the current flowing into M12 and raises the output trigger. V_{ref} can be adjusted externally in order to provide an output trigger signal only when the voltage difference between $V_{outpeak}$ and V_{inpeak} overcomes a suitable threshold (e.g. the noise level).

IV. Experimental Results

A first prototype of the proposed circuit has been realized in the 1μm/5V JFET/CMOS technology of the Fraunhofer-Institut für Mikroelektronische Schaltungen und Systeme, Duisburg. In this first prototype, a complete single-channel composed by input amplifier, four units (V/I converter and integration stage), peak detector, threshold discriminator and output line driver have been implemented. In Figure 6 a complete view of the circuit layout is shown.

In order to verify the basic functionality of the circuit and to evaluate the performances of the single stages, different

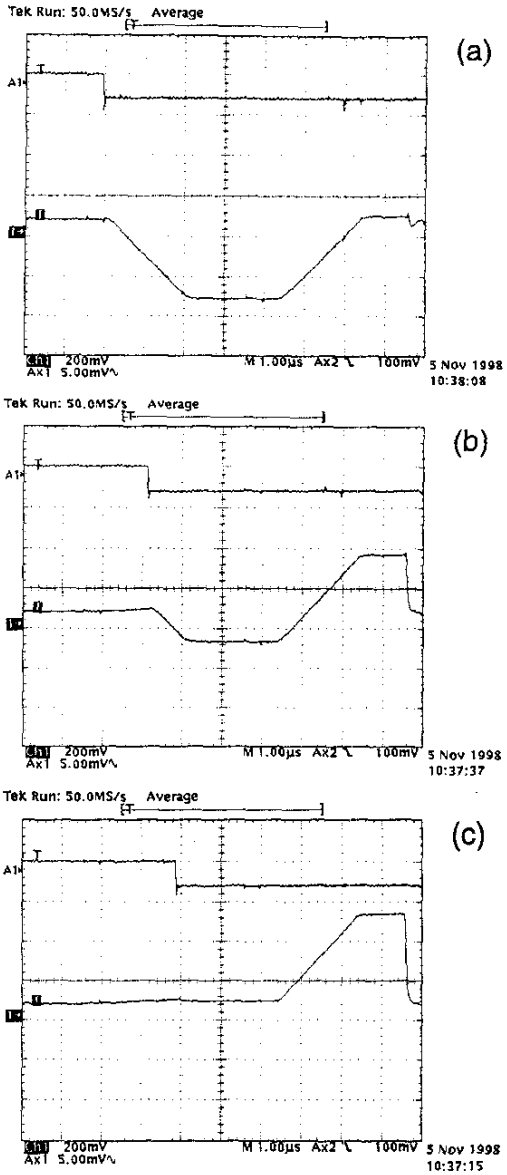

Figure 7: Integration sequence of one single unit for different arrival times of the input signal. The oscilloscope is triggered on the system clock. In the cases (b) and (c) the integration of an offset current before the signal occurrence can be noted.

timing sequences was properly generated by means of a programmable logic device (FPGA, Field Programmable Gate Array). The digital signals was provided for the operation of the switches through sinchronous latches, with a precision in the signal transistion time in the order of 1 ns.

Figure 7 shows the result of a typical double-integration phase at the output of one unit at three different time of occurrence of the input signal (shown in the upper side of each section of the Figure). In the case (a), the signal arrives before both integration phases, thus it is fully integrated in

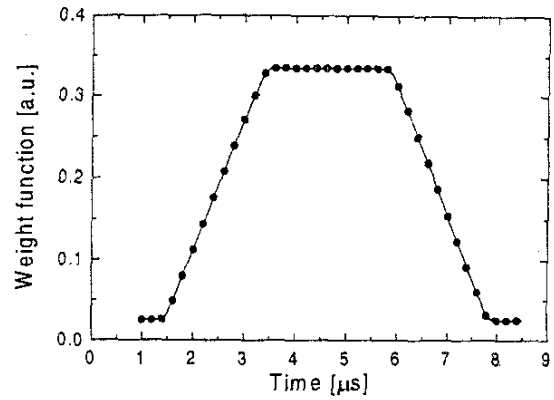

Figure 8: Weight function measured while setting at 2 µs the time of the integrations.

opposite directions (with an intermediate phase where it is sent to ground), leading to an almost zero net signal at the output of the unit. This result arises from an extremely good equality between the two integration phases. It can be noted that in this situation the weight given by the unit to the signal is zero. In the case (b), the signal occurs during the first integration phase, so it is partially integrated in the first phase and fully integrated in the second phase, leading to a net output signal which still represents the weight (non-zero in this case) given to the signal by the unit. In the last case (b), the signal arrives between the two integrations, thus it is fully integrated in the second phase and not in the first phase, leading to a net output signal higher than the previous one. In this case, the weight given by the unit to the signal is maximum, corresponding to a measurement performed during the flat-top of the weight function. From all measurements it can be noted that the very fast transitions between the integration phases and the intermediate phase where the signal is sent to ground indicates a satisfactory bandwidth of all circuits, employing operational amplifiers, used for the V/I convertion and integration phases. The weight function has been measured point by point (Figure 8) by varying the arrival time of the input step relatively to the timing of the unit.

The circuit has been than fully operated with all four units active and a suitable timing which guarantees that the input signal is processed with the maximum weight given by, at least, one unit. The output waveform consists of the sequence of the output values of the units, sampled by means of the S&H. In Figure 9 the output of the S&H is shown at three different times of occurrence of the input signal (shown again in the upper side of each section). The output waveform, which shows a "staircase shape", consists of the sequence of the different weights given by the units to the signal. Each weight is obtained for each unit as shown in Figure 7. It can be noted that, as expected, the maximum of the output signal is always the same irrespective of the occurrence time of the event.

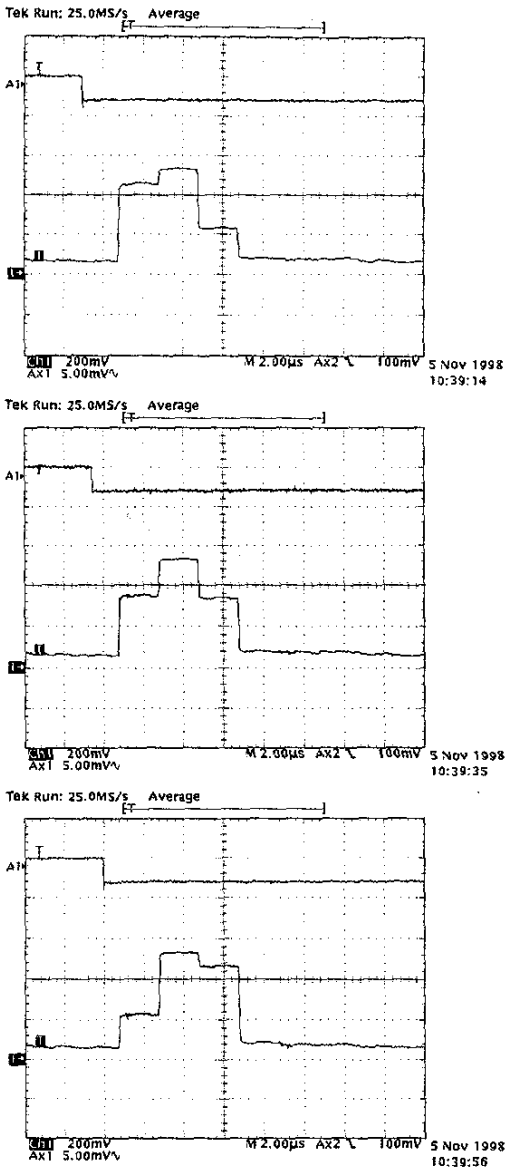

Figure 9: Output waveform of the Sample and Hold for three different times of arrival of the input signal.

The peak stretcher and threshold discriminator have been also operated in the same conditions and the output waveforms analog and digital, respectively, are shown in Figure 10 together with the input signal and the S&H output. The PS is correctly tracking the S&H output signal and then it holds the "staircase" maximum value. A droop rate higher than 10 µV/µs has been measured. The TD gives a negative digital signal when the S&H falls form its maximum value and the PS-S&H difference reaches a value higher than the threshold voltage set externally.

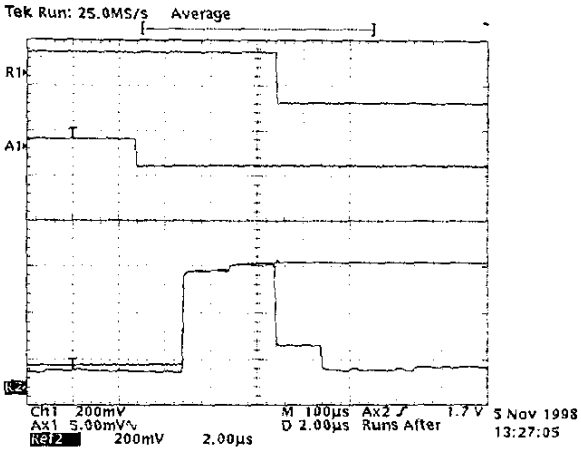

Figure 10: Output waveforms of the Threshold Discriminator, input signal, Peak Stretcher and S&H.

VI. References

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